Mirror Observerhood Lab IV

Minimal Reliability Thresholds in Viability-Constrained Agents

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Abstract

Earlier Observerhood Labs showed that reliability tracking can help, but only when it is action-guiding and cost-sensitive. Lab IV turns that claim into a threshold experiment. A non-repair baseline and a Mirror repair agent are compared across a sweep of self-state perturbation rate, repair cost and reliability threshold. Across 28,800 deterministic episodes, the positive region has the expected shape: Mirror-style repair is broadly beneficial when self-state perturbation is present and repair cost is low to moderate; the advantage becomes mixed at intermediate cost and disappears when repair becomes too expensive. The study gives computational form to the salience condition that self-model reliability is observer-relevant only where it improves expected viability after cost.

Keywords: Mirror Programme; Observerhood Labs; observerhood; recursive observerhood; self-model reliability; minimal reliability thresholds; viability; salience; perturbation; repair cost; threshold sweep; grid-world agents; computational cognition; symbolic artificial intelligence; constraint systems.

Contents

1 Roadmap position

This release forms part of the Computational Observerhood Labs of Mirror Programme, Volume I: Observerhood. The programme-level overview introduces a shared spine of constraint, model, reliability, repair, memory, viability, continuity and scale [1]. The Volume I overview applies that spine to observerhood as a derived organisational condition rather than a primitive subject or witness [2]. Mirror Theory I derives recursive observerhood from bounded viability-constrained organisation, including compressed self-models and reliability estimates [3]. Mirror Mathematics I supplies the first formal calculus for constraint regimes, local modelling, viability and reliability salience [4].

Lab I showed that scalar self-model reliability can improve viability under false self-location [5]. Lab II showed that decomposing reliability into self, sensor and map channels is informative but insufficient unless diagnosis is coupled to effective repair [6]. Lab III then showed that reliability becomes viability-relevant only when it is actionable and cost-sensitive [7]. Lab IV asks the threshold question: where, in a controlled parameter sweep, does reliability-gated repair become worth its cost?

2 Core claim

The core claim of Lab IV is conditional:

> A reliability-gated self-repair policy becomes observer-relevant only inside a bounded positive region where self-state perturbation is strong enough to threaten action and repair is cheap enough to improve expected viability after cost.

The negative side is equally important. Reliability tracking is not a monotonic good. Repair can become neutral where there is no self-state perturbation, and harmful where repair cost exceeds avoided viability loss. The point of the experiment is therefore not to show universal superiority, but to identify the boundary of positive salience.

3 Scope and negative commitments

This study is a controlled computational experiment, not a benchmark for general intelligence or consciousness. The environment is deliberately small. The reliability estimator and repair policy are hand-coded. The threshold sweep is an ablation over a toy viability functional, not a claim about biological cognition or physical observerhood.

The paper makes five negative commitments:

1. It does not claim that reliability thresholds are sufficient for observerhood.
2. It does not claim that Mirror repair is always superior to a baseline policy.
3. It does not claim that the best threshold can be known in advance by a deployed agent.
4. It does not treat the grid-world viability score as a universal value function.
5. It does not rely on unpublished mathematics papers as prior authority; it provides computational scaffolding for the later formal threshold synthesis.

4 Formal framing

The relevant salience expression is

$$\Sigma_L(R_L) = \mathbb{E}[V_L \mid M_E, M_L, R_L] - \mathbb{E}[V_L \mid M_E, M_L] - K_L(R_L),$$

where M_E is the environmental model, M_L is the local self-model, R_L is reliability information over that self-model and $K_L(R_L)$ is the modelling and repair cost associated with using reliability.

In this experiment, salience depends on three controlled variables: self-state perturbation rate p, repair cost c and reliability threshold τ. The Mirror agent repairs when its reliability estimate falls below threshold and when the estimated avoided loss exceeds repair cost. The baseline agent has the same environment but no repair policy.

Prediction 1 (Threshold prediction)**.** *Mirror-style reliability repair should show a positive region when self-state perturbation is non-zero and repair is cheap enough, and should lose advantage when repair cost exceeds avoided viability loss.*

5 Experimental design

5.1 Environment

The experiment uses a minimal grid-world survival task. The agent begins at a fixed start state and attempts to reach a goal while avoiding hazards and collecting energy packs. Self-state perturbation corrupts the represented self-location rather than the physical state itself. This isolates the question of whether reliability over the self-model becomes useful when action is selected from a corrupted self-representation.

The task-local viability score combines progress, energy, survival, success and hazard avoidance:

$$V = S + 0.75T + 0.65E - 9H + 30G + 15A,$$

where S contains goal and resource score, T is episode length, E is final energy, H is hazard hits, G indicates goal success and A indicates survival. Repair cost is subtracted from score and energy when the Mirror agent repairs.

This score is not a general theory of value. It is an explicit operational viability score for the controlled environment.

5.2 Parameter sweep

The sweep varies:

- eight self-state perturbation rates: $0.00, 0.04, 0.08, 0.12, 0.16, 0.20, 0.25, 0.30$;
- eight repair costs: $0, 2, 4, 6, 8, 10, 12, 15$;
- four reliability thresholds: $0.35, 0.50, 0.65, 0.80$.

For each perturbation-cost cell, the baseline is run once per seed and the Mirror agent is run at each threshold. Common random numbers are used across variants so that comparisons are architectural rather than stochastic.

5.3 Episodes and reproducibility

The experiment uses 90 episodes per perturbation-cost-threshold cell. Across eight perturbation rates, eight repair costs, one baseline and four Mirror thresholds, this produces 28,800 deterministic episodes. The accompanying reproducibility package contains the single-file Python implementation, fixed-seed data, summary tables and figure-generation code.

6 Results

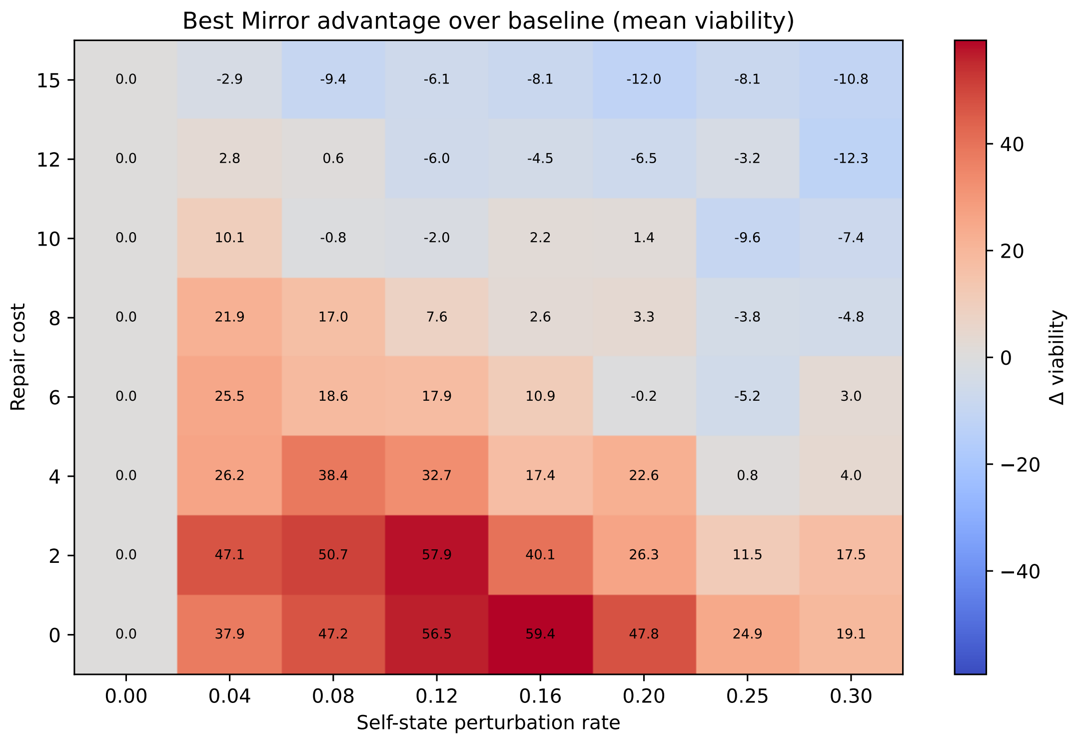

Figure 1: Best Mirror advantage over the baseline across perturbation and repair-cost cells. The positive region is broad at low repair cost and shrinks as cost rises.

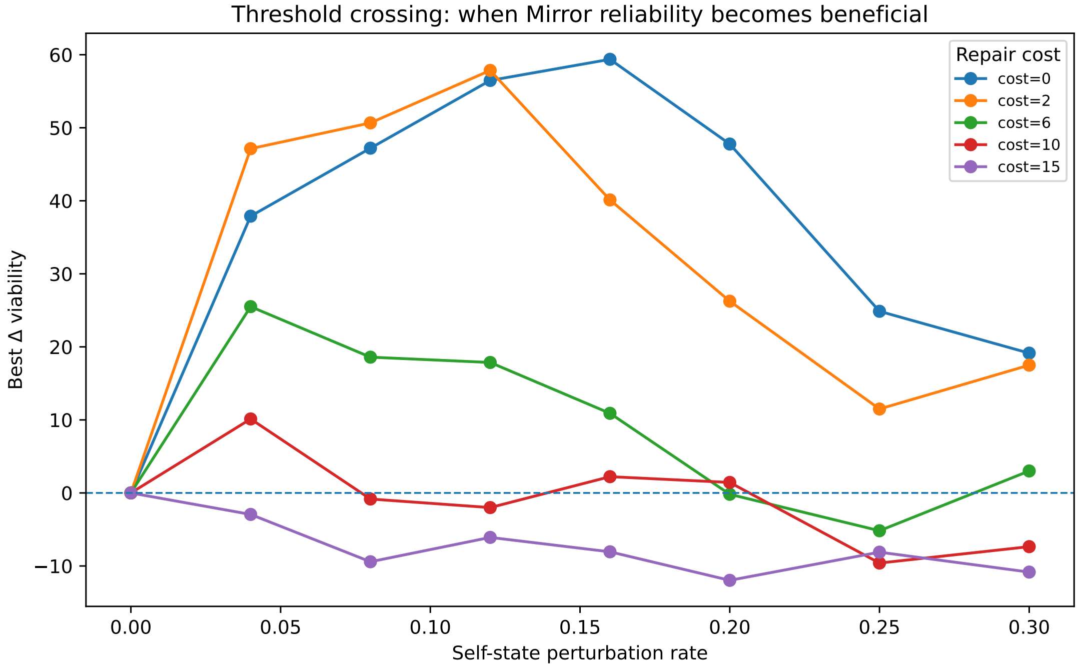

Figure 2: Threshold crossing curves for selected repair costs. At low cost, self-state perturbation creates a large reliability advantage; at high cost, repair becomes maladaptive.

Table 1: Threshold summary by repair cost. Minimal positive perturbation rate is the first tested perturbation rate at which the best Mirror threshold outperforms the baseline.

Repair cost	Min positive rate	Max ΔV	Rate at max	Threshold at max
0	0.04	59.4	0.16	0.80
2	0.04	57.9	0.12	0.80
4	0.04	38.4	0.08	0.80
6	0.04	25.5	0.04	0.80
8	0.04	21.9	0.04	0.65
10	0.04	10.1	0.04	0.50
12	0.04	2.8	0.04	0.50
15	–	0.0	0.00	0.35

The phase diagram gives the expected signature of a cost-sensitive salience condition. At repair cost 0, the maximum observed Mirror advantage is 59.4 viability points at perturbation rate 0.16 with threshold 0.80. At repair cost 2, the maximum advantage is 57.9 at perturbation rate 0.12. At repair costs 4 through 12, the positive region shrinks and becomes more local. At repair cost 15, no positive perturbation rate is observed in the best-threshold summary.

The zero-perturbation column is flat. This is important: where no self-state perturbation is present, reliability-gated repair has no positive role to play. The positive region emerges only when the agent's represented self-location is sufficiently vulnerable and repair remains cheap enough to offset the avoided viability loss.

7 Interpretation

The threshold surface confirms that recursive self-reliability is not a monotonic good. Its value depends jointly on perturbation intensity, repair cost and threshold placement. This is precisely the point of a salience functional rather than a simple architectural preference.

The experiment also shows why the Mathematics Arc needs a threshold predicate rather than a merely descriptive reliability variable. A reliability estimate becomes observer-relevant only where it changes repair or commitment in a way that improves expected viability after cost. In low-cost regions, repair can preserve action from corrupted self-location. In high-cost regions, even correct diagnosis can reduce viability if the intervention is too expensive.

8 Relationship to the Mathematics Arc

Lab IV turns the earlier computational findings into a threshold structure. Labs I through III establish that reliability must be self-relevant, decomposed enough to support diagnosis, actionable and cost-sensitive. Lab IV adds the parameter-sweep result: those conditions define a bounded positive region rather than a universal advantage.

This makes Lab IV the direct computational precursor to Mirror Mathematics II. The present paper does not cite that later mathematics paper as an already-published authority. Instead, it supplies the empirical scaffold for a subsequent formalisation of minimal observer thresholds.

9 Limitations

The limitations are substantive.

First, the best-threshold analysis is optimistic because it selects the best threshold within each perturbation-cost cell. A deployed agent would need to learn, infer or adapt thresholds online.

Second, the repair action is idealised. It restores represented self-location directly. Richer agents would require more realistic diagnostic and repair policies.

Third, the environment is small and fixed. The hazard map, energy packs and goal structure are intentionally simple so that the parameter sweep remains interpretable.

Fourth, the viability functional is task-local. It should not be read as a universal measure of value, intelligence, consciousness or observerhood.

Fifth, the result is computational scaffolding. It identifies a threshold-shaped condition under which reliability can matter; it does not by itself complete the formal theory.

10 Reproducibility statement

The source package accompanying this publication contains a single-file Python implementation of the environment, baseline agent, Mirror repair agent, parameter sweep, summary generation and figure generation. Running the default command

```
python mirror_lab_iv.py --episodes 90 --seed 4404 --outdir outputs/lab_iv
```

reproduces the 28,800-episode dataset, summary tables, all-threshold advantage table, best-threshold table, threshold summary and figures used in this paper.

11 Conclusion

Mirror-style self-reliability becomes beneficial in a bounded positive region. The result is not that reliability tracking is always useful. The result is that reliability becomes viability-positive when self-relevant perturbation threatens action and repair is affordable. Lab IV therefore supplies the computational threshold structure needed for the later minimal observer predicate: reliability matters where it improves expected viability after cost.

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